

CSC Resources for Scientific Computing

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CSC's HPC Resources

CSC's national HPC resources are free-of-charge for academic institutions, state research institutes, or the National Archives of Finland. The services discussed here are:

- HPC clusters
 - Puhti
 - Mahti
 - Roihu
 - LUMI
- Object storage
 - Allas
 - LUMI-O
- Cloud platform
 - cPouta
 - ePouta

From Laptop to LUMI

Bureaucracy

- Accounts
- Budgets
- Privileges

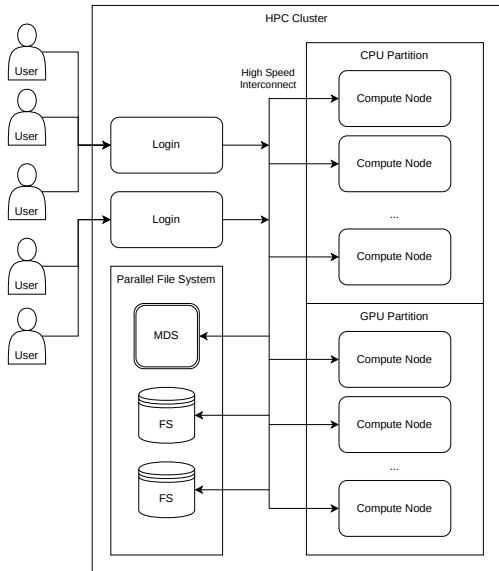
Technology

- Security
- Sharing resources
- Scheduling
- Software installation

Access

- CSC user account
 - Haka
- Computing project
 - Billing units (BUs), {C/G}PU hours
 - Managed by project PI
- Service

HPC Cluster Architecture



Cluster vs Laptop

Laptop

- Interactive
- Single version of software
- Few hard drives
- -
- GUI

Cluster

- Batch processing with slurm
- Module system
- Parallel file system
- Data management
- Command line*

Where to go

- Main website: www.csc.fi
- Services for research by field: research.csc.fi
- Technical documentation: docs.csc.fi
- Account management: my.csc.fi
- Web interface: e.g. mahti.csc.fi
- Tutorial for student projects:
https://docs.csc.fi/support/tutorials/student_quick/
- Weekly coffee break
<https://research.csc.fi/training/csc-research-support-coffee-every-wednesday-at-1400-finnish-time/>
- LUMI documentation: <https://docs.lumi-supercomputer.eu/>
- LUMI coffee break:
<https://lumi-supercomputer.eu/events/usercoffeebreaks/>

Accounting

Select an authentication provider

Which authentication provider should I use? ☒



Haka



CSC



Virtu

Accounting

CSC MFA Authentication

Please enter MFA code

Insert code from authenticator

Accounting

 Dashboard

 Projects

 Profile

 Resource Calculator

 Service Catalog

 Software Catalog

SSH PUBLIC KEYS

[+ Add key](#)



hpc-ssh-key

256 bits ed25519



Accounting

 Dashboard

 Projects

 Profile

 Resource Calculator

 Service Catalog

 Software Catalog

Roihu: Certificates



Slurm